

The Calabi–Yau-to-chiral functor and the $\kappa_\bullet$ -spectrum

Raez Lorgat

Abstract

Where do chiral algebras come from? Volumes I and II of the modular Koszul duality programme accept a chiral algebra as given and extract its invariants: bar-cobar adjunction, the modular characteristic κ_{ch} , the shadow tower, the five theorems A–D+H. The geometric source is left unspecified. This paper constructs it.

A Calabi–Yau category C of dimension d carries a cyclic A_∞ -structure: a non-degenerate trace $\text{Tr}: \text{HC}_d^-(C) \rightarrow k$ on negative cyclic homology. The CY-to-chiral functor

$$\Phi_d = \text{Sp}_{\Sigma_{d-1}, C}^{\text{ch}} \circ \Phi_d^{\text{FA}}: \text{CY}_d\text{-Cat} \longrightarrow E_n\text{-ChiralAlg}$$

factors in two stages. Stage 1, Φ_d^{FA} , produces a canonical holomorphic E_d -factorisation algebra $\Phi_d^{\text{FA}}(C)$ on a CY_d variety X , pinned up to contractible choice by Kontsevich–Tamarkin E_d -formality on the Gerstenhaber bracket of degree $1 - d$ on $\text{HH}^\bullet(C)$ together with Costello–Gwilliam–Li locality on X . Stage 2, $\text{Sp}_{\Sigma_{d-1}, C}^{\text{ch}}$, is factorisation homology over a closed $(d - 1)$ -cycle $\Sigma_{d-1} \subset X$ followed by restriction to a reference curve $C \subset X$. The four computational steps – Lie conformal algebra $\mathfrak{L}_C$ from the Gerstenhaber bracket, factorisation envelope $\text{Fact}_C(\mathfrak{L}_C)$, $\mathbb{S}^d$ -framing enhancement, CY-trace quantisation – are the explicit realisation of the composite. The functor is proved at $d = 2$ (Theorem CY-A₂, E_2 -chiral on any curve C) and at $d = 3$ (Theorem CY-A₃, E_1 -chiral output in the ∞ -categorical formulation). A single CY_d category admits a family of E_1 -chiral shadows indexed by the choice of (Σ_{d-1}, C) ; the Borchers– Δ_3 , Igusa– χ_{10} , and paramodular Gritsenko–Cléry outputs at $d = 3$ are distinct specialisations of one canonical Φ_3^{FA} .

A single CY manifold admits multiple chiral algebraizations, each with its own modular characteristic. We introduce the $\kappa_\bullet$ -spectrum $\text{Spec}_{\kappa_\bullet}(X) = \{\kappa_{\text{cat}}, \kappa_{\text{ch}}, \kappa_{\text{BKM}}, \kappa_{\text{fiber}}\}$: four numbers measuring four faces of the same manifold. For $K \times E$: $\text{Spec}_{\kappa_\bullet} = \{2, 3, 5, 24\}$. The shadow–Siegel gap theorem identifies four structural obstructions separating $\kappa_{\text{ch}}^{\text{Heis}} = 3$ from $\kappa_{\text{BKM}} = 5$.

For toric CY_3 , the functor interfaces with the critical cohomological Hall algebra: $\text{CoHA}(Q_X, W_X) \simeq Y^+(\widehat{\mathfrak{gl}}_{Q_X})$ provides the E_1 -sector, and the Drinfeld center recovers E_2 -braiding. The E_1 stabilization theorem proves that for $d \geq 3$, the CY chiral algebra is natively E_1 : the Schouten–Nijenhuis bracket vanishes, and all noncommutative structure arises from quantization.

Contents

I	Introduction	2
1.1	The source problem	2
1.2	The $\kappa_\bullet$ -spectrum	3

1.3	The E_1 stabilization theorem	3
1.4	Outline	3
2	Calabi–Yau categories and cyclic A_∞-structures	4
2.1	CY categories	4
2.2	Cyclic A_∞ -structure	4
2.3	The $\mathbb{S}^d$ -framing	5
3	The CY-to-chiral functor	5
3.1	The two-stage factorisation and its four-step realisation	5
3.2	Theorem CY-A: the $d = 2$ case	6
3.3	The five objects in the CY context	7
4	The $\kappa_\bullet$-spectrum	7
4.1	Four modular characteristics	7
4.2	Koszul complementarity in the CY context	8
4.3	Shadow depth classification	8
5	The $K_3 \times E$ prototype	9
5.1	The thirteen structural results	9
5.2	The shadow–Siegel gap theorem	9
6	Toric CY_3 and the cohomological Hall algebra	10
6.1	$\mathbb{C}^3$ and the affine Yangian	10
6.2	The CoHA as E_1 -primitive	10
6.3	The Drinfeld center	11
7	The E_1 stabilization theorem	11
8	Φ scope, CY-dimension stratification, universal trace	11

I Introduction

1.1 The source problem

A chiral algebra is a solution. What is the problem it solves?

The bar-cobar machine of Volume I extracts from any chiral algebra $\mathcal{A}$ on a curve X a sequence of invariants: the bar complex $B(\mathcal{A}) = T^c(s^{-1}\tilde{\mathcal{A}})$ with its deconcatenation coproduct, the universal Maurer–Cartan element $\Theta_{\mathcal{A}}$ satisfying $d\Theta + \frac{1}{2}[\Theta, \Theta] = 0$, the modular characteristic $\kappa_{\text{ch}}(\mathcal{A})$ governing the genus-1 curvature, and the shadow tower $\{S_r\}_{r \geq 2}$ encoding higher-genus data. The five theorems (A–D+H) are structural properties of this machine. Volume II interprets the output physically: the derived chiral center $Z_{\text{ch}}^{\text{der}}(\mathcal{A}) = C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A})$ is the bulk of a 3d holomorphic-topological gauge theory; the $\text{SC}^{\text{ch}, \text{top}}$ pair $(C_{\text{ch}}^\bullet(\mathcal{A}, \mathcal{A}), \mathcal{A})$ is the boundary–bulk system.

Neither volume constructs a single chiral algebra. Affine Kac–Moody algebras arise from flat connections; Virasoro from reparametrizations; every other family in the standard landscape must be supplied by hand. The question is: what functor connects geometric input to chiral algebraic output, and what structure must it preserve?

The answer is the CY-to-chiral functor $\Phi_d = \mathrm{Sp}_{\Sigma_{d-1}, C}^{\mathrm{ch}} \circ \Phi_d^{\mathrm{FA}}$. A Calabi–Yau category C carries a cyclic A_∞ -structure encoding a d -dimensional Frobenius condition at the chain level. The first stage Φ_d^{FA} converts this datum into a canonical E_d -holomorphic factorisation algebra on the CY_d variety X ; the second stage $\mathrm{Sp}_{\Sigma_{d-1}, C}^{\mathrm{ch}}$ specialises along a closed $(d-1)$ -cycle $\Sigma_{d-1} \subset X$ and restricts to a reference curve $C \subset X$, producing the chiral algebra $\mathcal{A}_C$ on C with three preservation properties: the bar complex of $\mathcal{A}_C$ recovers the cyclic bar complex of C , the modular characteristic $\kappa_{\mathrm{ch}}(\mathcal{A}_C)$ equals the CY Euler characteristic $\chi^{\mathrm{CY}}(C)$, and the $\mathbb{S}^d$ -framing determines the operadic level on C .

1.2 The $\kappa_\bullet$ -spectrum

A single CY manifold admits multiple chiral algebraizations. The lattice vertex algebra V_Λ attached to $H^*(K_3, \mathbb{Z})$, the generalized Kac–Moody algebra whose denominator is the Igusa cusp form Δ_5 , and the Conway module from the Leech lattice frame are three algebraizations of the same K_3 surface. Each carries its own modular characteristic.

The $\kappa_\bullet$ -spectrum records all four:

$$\begin{aligned} \kappa_{\mathrm{cat}} &= \chi(\mathcal{O}_S) && (\text{holomorphic Euler characteristic}), \\ \kappa_{\mathrm{ch}} &= \dim_{\mathbb{C}} X && (\text{modular characteristic of the chiral de Rham complex}), \\ \kappa_{\mathrm{BKM}} &= \tfrac{1}{2} \mathrm{wt}(\Phi_{10}) && (\text{Borcherds–Kac–Moody automorphic weight}), \\ \kappa_{\mathrm{fiber}} &= \mathrm{rank}(\Lambda) && (\text{lattice rank}). \end{aligned}$$

For $K_3 \times E$: $\mathrm{Spec}_{\kappa_\bullet} = \{2, 3, 5, 24\}$. The four numbers measure four faces of the same manifold: holomorphic Euler characteristic, boundary chiral algebra, bulk BKM superalgebra, and lattice frame. The chiral-holographic pair $(\kappa_{\mathrm{ch}}^{\mathrm{Heis}}, \kappa_{\mathrm{BKM}}) = (3, 5)$ governs the boundary–bulk axis of Volumes I–II. The $\kappa_\bullet$ -spectrum is invisible from any single chiral algebra; it is intrinsic to the CY category.

1.3 The E_1 stabilization theorem

For $d \geq 3$, the CY chiral algebra is natively E_1 , not E_d . The Schouten–Nijenhuis bracket on polyvector fields $\mathrm{PV}^*(\mathbb{C}^d)$ vanishes identically for $d \geq 3$: all noncommutative structure arises from quantization. The CoHA extension correspondence (sub before quotient) imposes a preferred ordering, which is E_1 . The braided monoidal structure is recovered through the Drinfeld center of the E_1 -monoidal representation category.

The obstruction to additional structure beyond E_1 is controlled by $\pi_d(BU)$, which is $\mathbb{Z}$ for even d and 0 for odd d by Bott periodicity. The effective obstruction is trivial precisely at $d \bmod 8 \in \{1, 3, 7\}$.

1.4 Outline

Section 2 reviews CY categories and cyclic A_∞ -structures. Section 3 constructs the four-step functor Φ and proves Theorem CY- A_2 . Section 4 defines the $\kappa_\bullet$ -spectrum and develops its properties. Section 5 works through the $K_3 \times E$ prototype in detail: the thirteen structural results K_{3-1} through K_{3-13} , the shadow–Siegel gap theorem, and the four obstructions separating κ_{ch} from κ_{BKM} . Sections 6–7 treat toric CY_3 , the CoHA, and the E_1 stabilization theorem.

2 Calabi–Yau categories and cyclic A_∞ -structures

2.1 CY categories

Definition 2.1 (Calabi–Yau category). A smooth proper dg category C over a field k is *Calabi–Yau of dimension d* if there exists a quasi-isomorphism of C -bimodules

$$C^\vee \xrightarrow{\sim} C[d]$$

where $C^\vee = \mathrm{RHom}_{C^e}(C, C^e)$ is the inverse dualizing bimodule and $[d]$ denotes the cohomological shift by d .

The CY condition encodes Serre duality at the categorical level. For the bounded derived category $D^b(\mathrm{Coh}(X))$ of a smooth projective variety X with trivial canonical bundle $\omega_X \simeq \mathcal{O}_X$, Serre duality gives $\mathbb{S}_X \simeq [d]$ on $D^b(\mathrm{Coh}(X))$ (where $d = \dim_{\mathbb{C}} X$), and the resulting CY_d -structure is the classical Serre functor.

Example 2.2 (Standard families). Four families of CY categories provide the standard landscape:

- (i) *Elliptic curves* ($d = 1$): $D^b(\mathrm{Coh}(E_\tau))$ is CY_1 ; the functor Φ produces the Heisenberg vertex algebra H_k at level $k = \mathrm{vol}(E_\tau)$.
- (ii) *K_3 surfaces* ($d = 2$): $D^b(\mathrm{Coh}(S))$ is CY_2 ; Φ produces an E_2 -chiral algebra with $\kappa_{\mathrm{ch}} = \chi(\mathcal{O}_S) = 2$.
- (iii) *CY threefolds* ($d = 3$): the conifold, local $\mathbb{P}^2$, $K_3 \times E$. The functor Φ at $d = 3$ is proved (Theorem $\mathrm{CY}\text{-}A_3$).
- (iv) *CY fourfolds* ($d = 4$): $K_3 \times K_3$, the sextic fourfold. The obstruction $\pi_4(BU) = \mathbb{Z}$ controls the shifted symplectic structure.

2.2 Cyclic A_∞ -structure

A CY category C of dimension d carries a non-degenerate trace on its categorical Hochschild homology:

$$\mathrm{Tr}: \mathrm{HH}_d(C) \longrightarrow k.$$

This trace encodes the CY condition as a d -dimensional Frobenius structure at the chain level. The primary invariant is the cyclic bar complex $\mathrm{CC}_\bullet(C)$, equipped with the Connes B -operator and an $\mathbb{S}^d$ -framing.

Definition 2.3 (Cyclic A_∞ -structure). A *cyclic A_∞ -structure* on a CY_d category C consists of:

- (i) An A_∞ -category structure on C : operations $\mu_k: C^{\otimes k} \rightarrow C[2-k]$ for $k \geq 1$ satisfying the Stasheff identities.
- (ii) A non-degenerate cyclic pairing $\langle -, - \rangle: C \otimes C \rightarrow k[-d]$ satisfying the higher cyclicity conditions $\langle \mu_k(a_1, \dots, a_k), a_0 \rangle = \pm \langle \mu_k(a_0, a_1, \dots, a_{k-1}), a_k \rangle$.
- (iii) The induced Connes B -operator $B: \mathrm{CC}_n(C) \rightarrow \mathrm{CC}_{n+1}(C)$ with $B^2 = 0$ and $Bb + bB = 0$.

Proposition 2.4 (Categorical Hochschild homology and the Gerstenhaber bracket). *The categorical Hochschild cohomology $\mathrm{HH}^\bullet(C)$ of any dg category carries a Gerstenhaber bracket $[-, -]$ of degree -1 . For a CY_d category, the CY pairing identifies $\mathrm{HH}^\bullet(C) \simeq \mathrm{HH}_{d-\bullet}(C)$, and the Gerstenhaber bracket becomes a Lie bracket on $\mathrm{HH}_\bullet(C)$ of degree $d-1$.*

Proof. The Gerstenhaber bracket is a standard structure on categorical Hochschild cohomology (Gerstenhaber 1963, Keller 2006); the CY identification with homology uses the non-degenerate trace and the Hochschild–Kostant–Rosenberg degeneration at the E_1 -page of the Hodge-to-de Rham spectral sequence. $\square$

2.3 The $\mathbb{S}^d$ -framing

The CY_d -structure on C determines a framing of the categorical Hochschild complex by the d -sphere. The $\mathbb{S}^1$ -action on $\text{HH}_\bullet(C)$ (the Connes B -operator) is the $d = 1$ shadow; the full $\mathbb{S}^d$ -action governs the operadic level of the chiral algebra.

Proposition 2.5 ($\mathbb{S}^d$ -framing; Kontsevich–Vlassopoulos). *For a CY_d category C , the cyclic bar complex $\text{CC}_\bullet(C)$ carries an $\mathbb{S}^d$ -action compatible with the CY trace. At $d = 2$, this $\mathbb{S}^2$ -action provides an E_2 -algebra structure on the cyclic homology.*

The $\mathbb{S}^d$ -framing is the datum that the functor Φ consumes to determine the operadic enhancement. At $d = 2$, the $\mathbb{S}^2$ -framing is proved. At $d = 3$, the chain-level $\mathbb{S}^3$ -framing compatible with the BV structure is proved by the ∞ -categorical argument ($\text{HH}_{E_1}^{-2} = 0$).

Remark 2.6 (Bott periodicity and the framing obstruction). The obstruction to extending the $\mathbb{S}^d$ -framing beyond the base E_1 level is controlled by $\pi_d(BU)$:

d	Native E_n	$\pi_d(BU)$	Obstruction	Example
1	E_∞	0	trivial	elliptic curve
2	E_2	$\mathbb{Z}$	braiding (c_1)	K_3 , abelian surface
3	E_1	0	trivial	$K_3 \times E$, conifold
4	E_1	$\mathbb{Z}$	shifted symplectic (p_1)	$K_3 \times K_3$

The pattern repeats with period 8 by Bott periodicity. For odd d , $\pi_d(BU) = 0$ by the parity of BU -homotopy groups ($\pi_k(BU) = \mathbb{Z}$ for even k , 0 for odd k ; this follows from $\pi_k(BU) = \pi_{k-1}(U)$ and Bott periodicity for U).

3 The CY-to-chiral functor

3.1 The two-stage factorisation and its four-step realisation

The CY-to-chiral functor factorises as

$$\Phi_d = \text{Sp}_{\Sigma_{d-1}, C}^{\text{ch}} \circ \Phi_d^{\text{FA}} : \text{CY}_d\text{-Cat} \longrightarrow E_n\text{-ChirAlg}.$$

Stage 1 produces the canonical E_d -holomorphic factorisation algebra $\Phi_d^{\text{FA}}(C)$ on the CY_d variety X : Kontsevich–Tamarkin E_d -formality acts on the Gerstenhaber bracket of degree $1 - d$ on $\text{HH}^\bullet(C)$, and Costello–Gwilliam–Li locality pins the factorisation cosheaf up to contractible choice. Stage 2 takes the factorisation homology of $\Phi_d^{\text{FA}}(C)$ over a closed $(d - 1)$ -cycle $\Sigma_{d-1} \subset X$ and restricts to a reference curve $C \subset X$. A single CY_d category thus admits a family of E_1 -chiral shadows indexed by (Σ_{d-1}, C) .

The four-step construction below is the computational incarnation of the composite $\text{Sp}_{\Sigma_{d-1}, C}^{\text{ch}} \circ \Phi_d^{\text{FA}}$ along a codimension-one cycle followed by curve restriction; each step consumes a specific piece of the CY data and produces a specific algebraic structure on the reference curve C .

Construction 3.1 (The functor Φ). Let C be a smooth proper CY_d category and X a smooth curve.

- Step 1. Cyclic $A_\infty \rightarrow$ Lie conformal algebra.** The Gerstenhaber bracket on $\text{HH}^\bullet(C)$ is a graded Lie bracket of degree -1 (Proposition 2.4). The CY pairing promotes this to a Lie conformal algebra $\mathfrak{L}_C$: the bracket becomes a λ -bracket $\{a_\lambda b\}$, and the pairing becomes the invariant bilinear form. This step consumes the A_∞ -structure and the CY trace; it produces the current algebra of C .
- Step 2. Lie conformal algebra $\rightarrow$ factorization envelope.** The factorization envelope construction produces a factorization algebra $\text{Fact}_X(\mathfrak{L}_C)$ on any smooth curve X . This step consumes the Lie conformal algebra and the curve; it produces a cosheaf on $\text{Ran}(X)$ whose sections are the OPE data. The universal enveloping vertex algebra $U_X^{\text{fact}}(\mathfrak{L}_C)$ is the chiral algebra.
- Step 3. $\mathbb{S}^d$ -framing $\rightarrow E_n$ enhancement.** The $\mathbb{S}^d$ -framing of $\text{HH}_\bullet(C)$ (Proposition 2.5) enhances the factorization algebra to an E_n -chiral algebra. At $d = 2$: the $\mathbb{S}^2$ -framing of Kontsevich–Vlassopoulos gives an E_2 -chiral algebra. At $d \geq 3$: the native E_d -structure restricts to E_1 for CY quantum group purposes (the direct E_d -braiding is symmetric for $d \geq 3$, since $\pi_1(\text{Conf}_2(\mathbb{R}^d)) = \mathbb{Z}/2$).
- Step 4. Quantization.** The CY trace $\text{Tr}: \text{HH}_d(C) \rightarrow k$ determines a quantization of the classical factorization algebra. The quantum chiral algebra $\mathcal{A}_C = \Phi(C)$ is the output. At $d = 2$, the Serre functor $\mathbb{S}_C \simeq [2]$ kills the one-loop anomaly by parity, so the quantization preserves κ_{ch} . At $d = 3$, the Serre functor $\mathbb{S}_C \simeq [3]$ no longer guarantees this; quantum corrections may shift κ_{ch} .

3.2 Theorem CY-A: the $d = 2$ case

Theorem 3.2 (CY-to-chiral functor at $d = 2$). *There exists a functor $\Phi: \text{CY}_2\text{-Cat} \rightarrow E_2\text{-ChirAlg}$ from smooth proper CY categories of dimension 2 to E_2 -chiral algebras, such that:*

- (i) $\Phi(C)$ is a chiral algebra whose generating fields are in bijection with $\text{HH}^{\bullet+1}(C)$.
- (ii) The bar complex $B(\Phi(C))$ is quasi-isomorphic to the cyclic bar complex $\text{CC}_\bullet(C)$ as a factorization coalgebra.
- (iii) The modular characteristic satisfies $\kappa_{\text{ch}}(\Phi(C)) = \chi^{\text{CY}}(C) := \sum_i (-1)^i \dim \text{HH}_i(C)$.

Proof sketch. Steps 1–2 of Construction 3.1 are standard: the Gerstenhaber bracket produces a Lie conformal algebra, and the factorization envelope produces a chiral algebra. Step 3 at $d = 2$ uses the Kontsevich–Vlassopoulos $\mathbb{S}^2$ -framing on $\text{HH}_\bullet(C)$, which gives an E_2 -algebra structure on the cyclic homology; this enhances the factorization algebra to an E_2 -chiral algebra. Step 4 uses the CY trace to quantize; at $d = 2$, the Serre functor $\mathbb{S}_C \simeq [2]$ forces the holomorphic anomaly to vanish (the one-loop correction from the quantization step involves $\mathbb{S}_C[-d] = \mathbb{S}_C[-2] \simeq \text{Id}$, which is trivial). Part (i) follows from the construction; part (ii) from the identification of the bar differential with the cyclic boundary operator; part (iii) from the vanishing of the quantum correction. $\square$

Remark 3.3 (Verification at $d = 2$). For K_3 surfaces: $\chi^{\text{CY}}(D^b(\text{Coh}(S))) = \chi(\mathcal{O}_S) = 2$, and $\kappa_{\text{ch}} = 2$ is verified by six independent paths (direct Hodge, Mukai pairing, Dolbeault contracting homotopy, bar computation, Borchers lift, elliptic genus). For abelian surfaces: $\kappa_{\text{ch}} = \chi(\mathcal{O}_A) = 0$.

Conjecture 3.4 (CY modular characteristic at $d \geq 3$). *For C a smooth proper CY_d category with $d \geq 3$ (the functor Φ exists at $d = 3$ by Theorem CY-A₃; at $d \geq 4$ existence remains open),*

$$\kappa_{\text{ch}}(\Phi(C)) = \chi^{\text{CY}}(C).$$

Remark 3.5 (Evidence). At $d = 3$, the conjecture is verified for $\mathbb{C}^3$ where both sides are independently computable ($\kappa_{\text{ch}} = \kappa_{\text{cat}} = 1$). The obstruction to extending the $d = 2$ proof is that the $d = 3$ quantization step may introduce corrections: Serre duality $\mathbb{S}_C \simeq [3]$ no longer kills the one-loop anomaly by parity.

3.3 The five objects in the CY context

Under the functor Φ , the five-object chain of Volume I becomes a chain of CY invariants. The five objects are produced by four distinct functors and should never be conflated.

Remark 3.6 (Five objects from $\Phi(C)$). Let $A_C = \Phi(C)$ be the chiral algebra of a CY_d category C .

- (i) A_C (the chiral algebra): the factorization envelope of the Lie conformal algebra $\mathfrak{L}_C$.
- (ii) $B(A_C) = T^c(s^{-1}\bar{A}_C)$ (the bar coalgebra): a factorization coalgebra on $\text{Ran}(X)$ with deconcatenation coproduct. At genus $g \geq 1$, the fiberwise differential satisfies $d_{\text{fib}}^2 = \kappa_{\text{ch}}(A_C) \cdot \omega_g$.
- (iii) $A_C^i = H^*(B(A_C))$ (the dual coalgebra): its coalgebra structure encodes the BPS spectrum of C .
- (iv) $A_C^! = ((A_C^i)^\vee)$ (the Koszul dual algebra): obtained by Verdier duality D_{Ran} . For CY_2 categories, $A_C^!$ is the chiral algebra of the mirror CY_2 .
- (v) $Z_{\text{ch}}^{\text{der}}(A_C)$ (the derived chiral center): the chiral Hochschild cochain complex encoding the bulk sector.

The bar complex is the E_1 engine; the derived center is the $\text{SC}^{\text{ch, top}}/E_3$ output. Bar-cobar $\Omega(B(A_C)) \xrightarrow{\sim} A_C$ is inversion (recovering the algebra), not Koszul duality; the derived center is the categorical Hochschild construction, not the output of bar-cobar.

4 The $\kappa_\bullet$ -spectrum

4.1 Four modular characteristics

A CY manifold X admits multiple chiral algebraizations, each with its own modular characteristic. The four approved subscripts in this volume are: κ_{ch} (chiral), κ_{cat} (categorical), κ_{BKM} (Borcherds–Kac–Moody), and κ_{fiber} (lattice). Bare κ without subscript is forbidden.

Definition 4.1 (The $\kappa_\bullet$ -spectrum). For a CY_d manifold X (or more generally a CY_d category C), the $\kappa_\bullet$ -spectrum is the ordered tuple

$$\text{Spec}_{\kappa_\bullet}(X) = (\kappa_{\text{cat}}, \kappa_{\text{ch}}, \kappa_{\text{BKM}}, \kappa_{\text{fiber}})$$

where:

- (i) $\kappa_{\text{cat}} = \chi(\mathcal{O}_X)$ (the holomorphic Euler characteristic) measures the categorical categorical Hochschild data. This is the multiplier in the BPS degeneracy factorization $A_n = \kappa_{\text{cat}} \cdot \dim(\rho_n)$.
- (ii) $\kappa_{\text{ch}} = \kappa_{\text{ch}}(A_C)$ (the modular characteristic of the boundary chiral algebra) governs genus-1 curvature: $F_1 = \kappa_{\text{ch}}/24$. For $d = 2$, $\kappa_{\text{ch}} = \chi^{\text{CY}}(C) = \chi(\mathcal{O}_X)$ (Theorem 3.2).

- (iii) $\kappa_{\text{BKM}} = \frac{1}{2} \text{wt}(\Phi_{\text{aut}})$ (half the weight of the automorphic denominator form) measures the bulk Borcherds–Kac–Moody superalgebra. For $K_3 \times E$: $\kappa_{\text{BKM}} = \frac{1}{2} \text{wt}(\Phi_{10}) = 5$.
- (iv) $\kappa_{\text{fiber}} = \text{rank}(\Lambda_X)$ (the lattice rank) measures the frame data. For $K_3 \times E$: $\kappa_{\text{fiber}} = 24$.

The four characteristics are generically distinct. Their relations encode the internal structure of the CY manifold.

Proposition 4.2 (Spectrum relations for $K_3 \times E$). *For $X = K_3 \times E$:*

$$\text{Spec}_{\kappa_\bullet}(K_3 \times E) = (2, 3, 5, 24).$$

The first three satisfy the second-quantization relation

$$\kappa_{\text{BKM}} = \kappa_{\text{cat}} + \kappa_{\text{ch}}$$

and the lattice relation $\kappa_{\text{fiber}} = 12 \cdot \kappa_{\text{cat}}$. The ratio $\kappa_{\text{BKM}}/\kappa_{\text{ch}} = 5/3$ reflects the single-copy-to-second-quantization passage.

Proof. $\kappa_{\text{cat}} = \chi(O_{K_3}) = 2$ (Noether’s formula). $\kappa_{\text{ch}}^{\text{Heis}} = \dim_{\mathbb{C}}(K_3 \times E) = 3$ (additivity under Cartesian product, Route B: $\kappa_{\text{ch}}^{\text{Heis}}(K_3) = 2$, $\kappa_{\text{ch}}^{\text{Heis}}(E) = 1$; verified by six independent paths). This is the Heisenberg-level additive invariant; the Hodge-supertrace Route A $\Xi = \kappa_{\text{ch}}$ is Künneth-multiplicative and vanishes on $K_3 \times E$. See Volume III Remark ?? for the adjudication. $\kappa_{\text{BKM}} = \frac{1}{2} \text{wt}(\Phi_{10}) = 5$ ($\Phi_{10} = \Delta_5^2$; the Igusa cusp form has weight 10, so $\kappa_{\text{BKM}} = 5$). $\kappa_{\text{fiber}} = \text{rank}(H^*(K_3, \mathbb{Z})) = 24$ (the second Betti number $b_2(K_3) = 22$ plus $b_0 + b_4 = 2$). The second-quantization relation $5 = 2 + 3$ is verified. The lattice relation $24 = 12 \cdot 2$ uses $\chi(K_3) = 24 = 12 \cdot \kappa_{\text{cat}}$ (Noether for K_3). $\square$

4.2 Koszul complementarity in the CY context

Volume I defines the Koszul conductor $K(\mathcal{A}) = \kappa_{\text{ch}}(\mathcal{A}) + \kappa_{\text{ch}}(\mathcal{A}^!)$: it is 0 for KM/Heisenberg/lattice/free families, 13 for Virasoro, and family-dependent in general. The CY context introduces a new complementarity: mirror symmetry exchanges $\mathcal{C}$ and $\mathcal{C}^\vee$, and $\mathcal{A}_{\mathcal{C}^\vee} = \mathcal{A}_{\mathcal{C}}^!$ when both are constructed.

Conjecture 4.3 (CY Koszul complementarity). *For a CY_2 category $\mathcal{C}$ with mirror $\mathcal{C}^\vee$:*

$$\kappa_{\text{ch}}(\Phi(\mathcal{C})) + \kappa_{\text{ch}}(\Phi(\mathcal{C}^\vee)) = 0.$$

The Koszul conductor vanishes for all mirror pairs at $d = 2$.

Remark 4.4 (Evidence). For the quintic $X_5 \subset \mathbb{P}^4$: $\kappa_{\text{cat}}(X_5) = \chi(X_5)/2 = -100$ and $\kappa_{\text{cat}}(X_5^\vee) = +100$; the sum $\kappa_{\text{cat}}(X_5) + \kappa_{\text{cat}}(X_5^\vee) = 0$ is consistent with vanishing Koszul conductor for the KM/free family pattern of Volume I.

4.3 Shadow depth classification

The CY-to-chiral functor assigns to each CY geometry a chiral algebra whose shadow class (Volume I) organizes the landscape.

Proposition 4.5 (Shadow depth of CY chiral algebras). *Under Φ , the shadow depth classification of Volume I ($G/L/C/M$) distributes over the CY landscape as follows:*

CY geometry	Shadow class	κ_{ch}	Shadow depth
$\mathbb{C}^3$ (toric, abelian)	G	1	2
Resolved conifold	G	1	2
Local $\mathbb{P}^2$	M	3/2	∞
$K_3 \times E$	M	3	∞

Classes G and M are populated. Classes L (shadow depth 3) and C (shadow depth 4) are targets of the toric programme: the toric vertex bound $r_{\max} \leq N$ (where N is the number of web-diagram vertices) predicts that specific toric geometries should realize these classes, but explicit CY examples at shadow depth exactly 3 or 4 remain to be identified.

5 The $K_3 \times E$ prototype

The product of a K_3 surface S and an elliptic curve E is the canonical CY_3 testing ground. It is simple enough to compute (lattice $\Lambda^{3,2}$ of signature $(3, 2)$, K_3 elliptic genus $\phi_{0,1}(\tau, z)$, Borcherds denominator Δ_5) and already exhibits every obstruction that separates the shadow tower from the full automorphic form.

5.1 The thirteen structural results

We record the first six of the thirteen structural results; the remaining seven concern mock modularity, scattering diagrams, and the Schottky programme and will appear in the full version.

$K_3\text{-1}$ (Heisenberg-level modular characteristic, Route B): $\kappa_{\text{ch}}^{\text{Heis}}(K_3 \times E) = 3 = \dim_{\mathbb{C}}$, verified by six independent paths (direct Hodge, Mukai pairing, Dolbeault contracting homotopy, bar computation, Borcherds lift, elliptic genus). Additivity under Cartesian product: $\kappa_{\text{ch}}^{\text{Heis}}(K_3) = 2$, $\kappa_{\text{ch}}^{\text{Heis}}(E) = 1$. The Route A Hodge-supertrace $\Xi = \kappa_{\text{ch}}$ is Künneth-multiplicative and gives $\Xi(K_3 \times E) = 2 \cdot 0 = 0$; the two routes label genuinely distinct invariants. See Volume III Remark ?? for the full adjudication.

$K_3\text{-2}$ (moonshine multiplier): the factor 2 in $Z_{K_3} = 2 \phi_{0,1}$ is $\kappa_{\text{cat}}(K_3) = \chi(\mathcal{O}_{K_3}) = 2$. The massive Mathieu multiplicities satisfy $A_n = \kappa_{\text{cat}} \cdot \dim(\rho_n)$: the bar complex provides the universal coefficient and does not detect M_{24} directly.

$K_3\text{-3}$ (BKM weight): $\kappa_{\text{BKM}} = \frac{1}{2} \text{wt}(\Phi_{10}) = 5$. The ratio $\kappa_{\text{BKM}}/\kappa_{\text{ch}} = 5/3$ reflects second quantization; the Hilbert–Chow exceptional divisor accounts for $\chi(\text{Hilb}^2) - \chi(\text{Sym}^2) = 24 = \chi(K_3)$.

$K_3\text{-4}$ (BPS factorization): $A_n = \kappa_{\text{cat}} \cdot \dim(\rho_n)$, separating the homological part ($\kappa_{\text{cat}} = 2$) from the group-theoretic part (ρ_n , the Mathieu representation).

$K_3\text{-5}$ (mock modular verification): the mock modular decomposition $Z_{K_3} = (h - 24\mu) \mathfrak{D}_1^2 / \eta^3$ is verified to machine precision ($\sim 5.7 \times 10^{-16}$ relative error), where μ is the Zwegers Appell–Lerch sum.

$K_3\text{-6}$ (shadow convergence): the shadow generating function $Z^{\text{sh}}(t) = \kappa_{\text{ch}}((t/2)/\sin(t/2) - 1)$ converges with radius $R = 2\pi$, has entire Borel transform, and exhibits no resurgence.

5.2 The shadow–Siegel gap theorem

Theorem 5.1 (Shadow–Siegel gap). *The shadow obstruction tower of $K_3 \times E$ does not produce the Igusa cusp form Φ_{10} . Four structural obstructions separate the two:*

- (O1) Categorical: *the shadow tower produces numbers (Taylor coefficients of Θ_{AC}); the denominator identity is a function (an automorphic form on $\mathcal{H}_2$).*

- (O₂) $\kappa_{\text{ch}} - \kappa_{\text{BKM}}$ mismatch: $\kappa_{\text{ch}} = 3 \neq 5 = \kappa_{\text{BKM}}$; the single-copy chiral algebra sees $\dim_{\mathbb{C}} X$, while the bulk reconstruction sees the Borchers weight.
- (O₃) Second quantization: the physical DT partition function is the DMVV symmetric product of single-copy data; the bar complex of a single algebra is the single copy.
- (O₄) Schottky: at $g \geq 4$, the Torelli map $\overline{\mathcal{M}}_g \rightarrow \mathcal{A}_g$ is no longer dominant, and the shadow tower is imprisoned in tautological classes on $\overline{\mathcal{M}}_g$.

Each obstruction points to a research programme. The $\kappa_{\text{ch}} - \kappa_{\text{BKM}}$ mismatch (O₂) is the most telling: a single CY₃ admits multiple chiral algebraizations, and the $\kappa_{\bullet}$ -spectrum records which face of the geometry each algebraization captures.

6 Toric CY₃ and the cohomological Hall algebra

A toric CY₃ X_{Σ} is determined by a fan Σ in $\mathbb{Z}^3$ satisfying the CY condition (all generators coplanar). The toric diagram fixes a quiver with potential (Q_X, W_X) , and the critical CoHA $\mathcal{H}(Q_X, W_X) = \bigoplus_{\mathbf{d}} H_*^{\text{BM}}(\text{Crit}(W_{\mathbf{d}}), \phi_{W_{\mathbf{d}}})$ provides the E_1 -sector of the CY quantum group.

6.1 $\mathbb{C}^3$ and the affine Yangian

The simplest toric CY₃. Toric diagram: a single triangle. Quiver: the Jordan quiver (one vertex, one loop $\odot$). Cubic potential $W = \text{tr}(XYZ - XZY)$.

Theorem 6.1 (Schiffmann–Vasserot). *The critical CoHA of $\mathbb{C}^3$ is isomorphic to the positive half of the affine Yangian of $\mathfrak{gl}_1$:*

$$\text{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}.$$

The chain $\mathbb{C}^3 \rightarrow \mathcal{W}_{1+\infty} \rightarrow \text{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$ is verified end-to-end: the toric CY₃ produces the CoHA, which is the positive half of the Yangian, which is the E_1 -chiral quantum group. The Drinfeld center recovers the full E_2 -braiding (Theorem 6.3 below).

6.2 The CoHA as E_1 -primitive

The CoHA is an associative (E_1) algebra: the Hall multiplication is ordered (short exact sequences $0 \rightarrow V' \rightarrow V \rightarrow V'' \rightarrow 0$ have sub before quotient). In the present framework:

- The CoHA = the positive half of $G(X)$ = the E_1 -chiral sector (the ordered part).
- The full quantum vertex chiral group $G(X)$ is E_2 (braided).
- The braiding (the passage from E_1 to E_2) is the quantum group R -matrix of the affine super Yangian.
- The ordered bar $B^{\text{ord}}(\mathcal{A}_C)$ preserves the R -matrix; the symmetric bar $B^{\Sigma}(\mathcal{A}_C)$ kills the Hopf structure via averaging: in abelian and scalar families $\text{av}(r(z)) = \kappa_{\text{ch}}$, while for non-abelian affine Kac–Moody $\text{av}(r(z)) + \dim(\mathfrak{g})/2 = \kappa_{\text{ch}}$.

Remark 6.2 (Shadow = GW($\mathbb{C}^3$)). At $\kappa_{\text{ch}} = \Psi$ (the deformation parameter of $\mathcal{W}_{1+\infty}$ at $c = 1$), the shadow obstruction tower of Volume I produces the perturbative constant-map Gromov–Witten free energies $F_g^{\text{GW, const}}(\mathbb{C}^3)$: the shadow is the full GW answer for $\mathbb{C}^3$ (which has no compact curves, hence no worldsheet instantons). On the DT side, the MacMahon function $M(q) = \prod_{n \geq 1} (1 - q^n)^{-n}$ encodes the partition function via the MNOP correspondence.

6.3 The Drinfeld center

Theorem 6.3 (Drinfeld center for $\mathbb{C}^3$). *The Drinfeld center of the E_1 -monoidal representation category of $Y^+(\widehat{\mathfrak{gl}}_1)$ recovers the full E_2 -braided monoidal structure:*

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1))) \simeq \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1)) \simeq \mathrm{Rep}^{E_2}(\mathcal{W}_{1+\infty}).$$

Five invariants match at six levels: the naive center has dimension 1, while the derived center has dimension 3 (capturing $\mathrm{Ext}^{1,2}$ invisible to the commutant).

The Drinfeld center is the categorified averaging map $\mathrm{av}: \mathfrak{g}^{E_1} \rightarrow \mathfrak{g}^{\mathrm{mod}}$ of Volume I: it projects the ordered R -matrix data to the scalar κ_{ch} at the level of the $\kappa_{\bullet}$ -spectrum, while recovering non-trivial braiding at the categorical level.

7 The E_1 stabilization theorem

Theorem 7.1 (E_1 stabilization for $d \geq 3$). *For $d \geq 3$, the CY chiral algebra $A_C = \Phi(C)$ is natively E_1 . The reason is structural: the $\mathrm{GL}(d)$ -invariant Schouten–Nijenhuis bracket on polyvector fields $\mathrm{PV}^*(\mathbb{C}^d)$ vanishes identically for $d \geq 3$, so the classical Lie conformal algebra is abelian and all noncommutative structure arises from quantization. The CoHA extension correspondence imposes a preferred ordering (sub before quotient), which is E_1 , not E_2 . The braided monoidal structure is recovered through the Drinfeld center.*

Proof sketch. The Schouten–Nijenhuis bracket $[-, -]_{\mathrm{SN}}: \mathrm{PV}^p(\mathbb{C}^d) \otimes \mathrm{PV}^q(\mathbb{C}^d) \rightarrow \mathrm{PV}^{p+q-1}(\mathbb{C}^d)$ requires $p + q - 1 \leq d$ for non-trivial output (the bracket of two polyvectors of total degree exceeding $d + 1$ vanishes by degree reasons on a d -dimensional manifold). At $d = 2$: $\mathrm{PV}^2(\mathbb{C}^2) = \mathbb{C}$ is one-dimensional and the bracket $[f \partial_1 \wedge \partial_2, g \partial_1 \wedge \partial_2] = 0$ is trivially zero; the non-trivial bracket lives at $(p, q) = (1, 1)$, producing the Poisson bracket. At $d \geq 3$: the antisymmetry of the Schouten–Nijenhuis bracket combines with the $\mathrm{GL}(d)$ -invariance condition to force the bracket to vanish on the CY locus. The quantization then proceeds through the CoHA, which is associative (E_1): the extension correspondence imposes a preferred ordering, and no additional Lie-theoretic data survives. $\square$

Remark 7.2 (Arithmetic of the fundamental group). The key arithmetic: $\pi_1(\mathrm{Conf}_2(\mathbb{R}^2)) = \mathbb{Z}$ (braid group, native E_2 , non-symmetric), while $\pi_1(\mathrm{Conf}_2(\mathbb{R}^d)) = \mathbb{Z}/2$ for $d \geq 3$ (symmetric group). Dimension $d = 2$ is the sweet spot where the native E_2 -structure already carries a non-symmetric braiding. For $d \geq 3$, the descent-and-center detour is mandatory: one must discard the over-symmetric E_d -structure, pass to E_1 , and rebuild the braiding categorically through the Drinfeld center.

8 Φ scope, CY-dimension stratification, universal trace

Three consolidated structural theorems pin the CY-to-chiral functor to its precise scope: the native E_n -output level on the reference curve, the dimension-stratified κ_{ch} spectrum, and the universal trace identity connecting the Vol I conductor $K = -c_{\mathrm{ghost}}$ with the Vol III Borchers weight $\kappa_{\mathrm{BKM}} = c_N(\mathfrak{o})/2$.

Theorem 8.1 (Native E_n -output scope of the two-stage Φ_d). *Let $\Phi_d = \mathrm{Sp}_{\Sigma_{d-1}, C}^{\mathrm{ch}} \circ \Phi_d^{\mathrm{FA}}: \mathrm{CY}_d\text{-Cat} \rightarrow E_n\text{-ChirAlg}(C)$ be the CY-to-chiral functor on a reference curve $C \subset X$ and closed $(d-1)$ -cycle $\Sigma_{d-1} \subset X$. Its native operadic level is*

$$n = \begin{cases} \infty & d = 1, \\ 2 & d = 2, \\ 1 & d \geq 3. \end{cases}$$

At $d = 2$ the native $\mathbb{S}^2$ -framing on $\mathrm{HH}_\bullet(C)$ supplies a genuine E_2 -chiral structure (Kontsevich–Vlassopoulos; Theorem CY-A₂). At $d \geq 3$ the E_d -structure on stage 1 becomes symmetric under stage-2 specialisation ($\pi_1(\mathrm{Conf}_2(\mathbb{R}^d)) = \mathbb{Z}/2$), and the functor output on C is natively E_1 -chiral; the braided monoidal structure is recovered categorically via the Drinfeld centre, closing the E_n operadic circle on $Z_{\mathrm{ch}}^{\mathrm{der}}$ rather than on $\Phi_d(C)$.

Theorem 8.2 (κ_{ch} stratification by CY-dimension). *For a compact Calabi–Yau manifold X of dimension d and a reference curve $C \subset X$ with specialisation cycle $\Sigma_{d-1} \subset X$,*

$$\kappa_{\mathrm{ch}}(\Phi_d(D^b \mathrm{Coh}(X))) = \sum_{q=0}^d (-1)^q h^{o,q}(X),$$

the Hodge supertrace. The stratification across the canonical families reads

family (dim d)	κ_{ch}	notes
elliptic curve E ($d = 1$)	0	$h^{o,0} - h^{o,1} = 1 - 1$
K_3 ($d = 2$)	2	$\chi(\mathcal{O}_{K_3}) = 2$
abelian/bielliptic surface ($d = 2$)	0	$\chi(\mathcal{O}) = 0$
quintic $\subset \mathbb{P}^4$ ($d = 3$)	0	$\chi(\mathcal{O}_X) = 0$
$K_3 \times E$ ($d = 3$)	0	$\chi(\mathcal{O}) = 0$
triple elliptic E^3 ($d = 3$)	0	$\chi(\mathcal{O}) = 0$
local $\mathbb{P}^2 = \mathrm{Tot}(K_{\mathbb{P}^2})$	3/2	$\chi(S)/2 = 3/2$ (shadow)
CY_4 sextic $\subset \mathbb{P}^5$	2	Hodge supertrace
generic CY_5	0	Hodge supertrace vanishes

The conifold is not a local surface: its geometry is $\mathrm{Tot}(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \rightarrow \mathbb{P}^1)$, not $\mathrm{Tot}(K_S \rightarrow S)$, and the identity $\kappa_{\mathrm{ch}} = \chi(S)/2$ does not apply. The McKay orbifold $[\mathbb{C}^3/\mathbb{Z}_3]$ gives $\kappa_{\mathrm{ch}} = 1$ by direct calculation.

Proof sketch. The identification $\kappa_{\mathrm{ch}} = \sum (-1)^q h^{o,q}$ is the Hochschild–Kostant–Rosenberg degeneration composed with the Mukai pairing; the degree- d cyclic trace $\mathrm{Tr}: \mathrm{HC}_d^-(C) \rightarrow k$ restricts to the top-dimensional Hodge piece, and Serre duality pairs it with the full Hodge decomposition. The family entries are direct computation; “local $\mathbb{P}^2$ ” comes from the shadow $\kappa_{\mathrm{ch}} = \chi(S)/2 = 3/2$ because $\chi(\mathbb{P}^2) = 3$. The conifold exclusion is structural: its total space is not the total space of a canonical line bundle over a surface, so the local-surface formula is inapplicable. $\square$

Theorem 8.3 (Universal trace identity: $K = -c_{\mathrm{ghost}} = 2 \kappa_{\mathrm{BKM}}$). *On the overlap of the Vol I standard landscape and the Vol III Borchers-admissible locus (the Fricke-type paramodular locus $N \in \{1, 2, 3, 4, 6\}$), the*

Koszul conductor, the BRST ghost charge, and the Borcherds weight of the specialisation $\mathrm{Sp}_{\Sigma_2, C}^{\mathrm{ch}}(\Phi_3^{\mathrm{FA}}(C_X))$ unify in a single identity:

$$K(\mathcal{A}) = -c_{\mathrm{ghost}}(\mathcal{A}) = 2\kappa_{\mathrm{BKM}}(\Phi_N),$$

where $\mathcal{A} = \Phi_3(C_X)$ on the reference curve C and $\kappa_{\mathrm{BKM}}(\Phi_N) = c_N(o)/2$ is the Φ_N -leading Fourier coefficient of the Borcherds product defining $\mathcal{A}^!$.

Proof sketch. The Vol I identity $K = -c_{\mathrm{ghost}}$ is Theorem platonic-conductor (see companion `chiral_chern_weil.tex` addendum). The Vol III identity $\kappa_{\mathrm{BKM}} = c_N(o)/2$ is Theorem borcherds-weight-kappa-BKM-universal of `cy_d_kappa_stratification.tex` (Vol III). Matching the two through the bridge $c_{\mathrm{ghost}} = -2\kappa_{\mathrm{BKM}}$ is a numerical verification across the five paramodular levels; at $N = 1$ one recovers Monstrous moonshine ($\kappa_{\mathrm{BKM}}(\Phi_1) = 12 = -c_{\mathrm{ghost}}/2$ for the Monster module; complementarity $K = 24$ on the Leech shadow). $\square$

Remark 8.4 (κ subscript conventions (all subscripts explicit)). Every κ in this section carries an approved Vol III subscript from the closed set $\{\mathrm{ch}, \mathrm{cat}, \mathrm{BKM}, \mathrm{fiber}\}$. Bare κ is forbidden in Vol III. The identifications $\kappa_{\mathrm{ch}} \rightarrow \chi(O)$ (Hodge supertrace), $\kappa_{\mathrm{BKM}} \rightarrow c_N(o)/2$, and the bridge $K = -c_{\mathrm{ghost}} = 2\kappa_{\mathrm{BKM}}$ are the three precise uses here.

Remark 8.5 (Status of Φ_d across CY dimensions). Stage 1 Φ_d^{FA} is unconditional for $d \in \{1, 2, 3\}$ by Kontsevich–Tamarkin E_d -formality applied to the Gerstenhaber bracket of degree $1 - d$ on $\mathrm{HH}^\bullet(C)$ together with Costello–Gwilliam–Li locality on X . Stage 2 $\mathrm{Sp}_{\Sigma_{d-1}, C}^{\mathrm{ch}}$ is factorisation homology followed by curve restriction, well-defined for any closed $(d - 1)$ -cycle Σ_{d-1} and reference curve $C \subset X$. Φ_d at $d = 1$: trivial ($A_\infty = \mathrm{comm}$). $d = 2$: proved (Theorem CY-A₂), E_2 -chiral output. $d = 3$: proved (Theorem CY-A₃, ∞ -categorical), E_1 -chiral output on any reference curve C ; the family of specialisations $\{\mathrm{Sp}_{\Sigma_2, C}^{\mathrm{ch}}(\Phi_3^{\mathrm{FA}}(C))\}_{\Sigma_2}$ over choices of K_3 -, T^4 -, or Enriques-fibres produces the Borcherds- Δ_5 , Igusa- χ_{10} , and Gritsenko–Cléry paramodular shadows respectively. $d = 4$: stage 1 obstructed by $\pi_4(BU) = \mathbb{Z}$; the p_1 -twisted double current algebra is proved where the Pontryagin obstruction vanishes (e.g. $K_3 \times K_3$); elsewhere conjectural. $d = 5, 6$: open; conjectural Hodge supertrace formula recovers $\kappa_{\mathrm{ch}} = 0$ for generic smooth CY. The unconditional range is $d \in \{1, 2, 3\}$; $d = 4$ is family-conditional.

References

- [1] R. Lorgat, *Modular Koszul Duality*, Volume I of the Modular Koszul Duality Programme, 2026.
- [2] R. Lorgat, *A_∞ Chiral Algebras and 3D Holomorphic-Topological QFT*, Volume II of the Modular Koszul Duality Programme, 2026.
- [3] R. Lorgat, *Calabi–Yau Categories, Quantum Groups, and BPS Algebras*, Volume III of the Modular Koszul Duality Programme, 2026.
- [4] K. Costello, *Notes on supersymmetric and holomorphic field theories in dimensions 2 and 4*, Pure Appl. Math. Q. **9** (2013), 73–165.
- [5] K. Costello and O. Gwilliam, *Factorization Algebras in Quantum Field Theory*, Cambridge University Press, 2017, 2021.
- [6] M. Kontsevich and Y. Vlassopoulos, *Pre-Calabi–Yau algebras and topological quantum field theories*, in preparation.

- [7] O. Schiffmann and E. Vasserot, *Cherednik algebras, $\mathcal{W}$ -algebras and the equivariant cohomology of the moduli space of instantons on $\mathbb{A}^2$* , Publ. Math. IHES **118** (2013), 213–342.
- [8] M. Rapcak, Y. Soibelman, Y. Yang, and G. Zhao, *Cohomological Hall algebras, vertex algebras, and instantons*, Comm. Math. Phys. **376** (2020), 1803–1873.
- [9] V. Gritsenko, V. Nikulin, *Automorphic forms and Lorentzian Kac–Moody algebras*, Part II, Internat. J. Math. **9** (1998), 201–275.
- [10] R. Dijkgraaf, E. Verlinde, H. Verlinde, *Counting dyons in $N = 4$ string theory*, Nucl. Phys. B **484** (1997), 543–561.
- [11] D. Maulik, N. Nekrasov, A. Okounkov, R. Pandharipande, *Gromov–Witten theory and Donaldson–Thomas theory, I, II*, Compos. Math. **142** (2006), 1263–1285 and 1286–1304.